BDS-FAQ-001

Staff Interview FAQ Document

Benamdaj Dredging Solutions Ltd.

56 Q&A Pairs | 6 Staff Personas | 8 Topic Categories | 4 Dredger Models

Field	Detail
Document Reference	BDS-FAQ-001
Document Title	Staff Interview FAQ Document
Company	Benamdaj Dredging Solutions Ltd. (RC: 1653951)
Purpose	Knowledge base component for Hybrid RAG customer support system
Knowledge Type	Tacit — elicited through structured staff interviews
Chunking Method	Q&A-pair-based (each Q&A = one retrieval chunk)
Total Q&A Pairs	56
Staff Personas	6 (roles spanning operator, technician, supervisor)
Topic Categories	8
Dredger Models Covered	All four models + fleet-wide content
Prepared for	MSc AI & Data Science Dissertation, University of Wolverhampton
Version	2.0 (revised — plain format, no company branding)

1. Document Overview

This document presents 56 question-and-answer pairs derived from structured interviews conducted with six members of staff at Benamdaj Dredging Solutions Ltd. The interviews were carried out as part of a research project investigating the design of a Hybrid Retrieval-Augmented Generation (RAG) customer support system for the company.

The primary purpose of this document is to serve as a knowledge base component for the RAG system. The Q&A pairs capture tacit operational knowledge — the practical experience, diagnostic judgement, and contextual understanding that experienced staff have developed through years of working with the company's four dredger models — and encode it in a structured, machine-readable format suitable for retrieval.

Each Q&A pair is tagged with the following metadata: the staff persona who provided the answer, the dredger model or models to which the answer applies, the topic category, and the retrieval priority. This metadata enables the RAG system to filter and rank results appropriately when responding to customer queries.

1.1 The Four Dredger Models

Model	Name	Main Engine	Drive	Pump Size	Max Depth
Model 1	16/14" Cutter Suction Dredger	Weichai X6170ZC650-2 (478 kW)	Marine gearbox	16/14"	22 m
Model 2	14/12" Amphibious Multifunctional	Weichai X6170ZC650-2 (478 kW)	Marine gearbox	14/12"	22 m
Model 3	12/10" Bucket Chain Dredger	Caterpillar 3408	Marine gearbox	12/10"	22 m
Model 4	10/10" Jet Suction Dredger	Weichai WP13C550-18E121 (405 kW)	PTO	10/10"	18 m

1.2 Knowledge Collection Methodology

The Q&A pairs in this document were collected through structured interviews with six members of staff across different roles and levels of experience. Each interview followed a topic guide covering eight subject areas relevant to customer support enquiries. Staff were asked to respond as they would if a customer or new colleague had asked the question — drawing on their practical knowledge rather than formal documentation.

The interview data captures knowledge that does not appear in the formal technical manuals. Senior staff articulate diagnostic heuristics, informal rules of thumb, and experiential understanding of equipment behaviour under real operating conditions. This tacit knowledge is particularly valuable for the RAG system because it closely mirrors the language that customers use when asking questions — colloquial, problem-focused, and experience-based rather than specification-driven.

The interviews were audio-recorded with the consent of each participant, transcribed, and structured into Q&A format by the researcher. Each answer has been lightly edited for clarity and readability whilst preserving the original meaning and register of the interviewee's response.

2. Staff Personas

Six staff members participated in the interview process. Their profiles are summarised below. Names have been replaced with persona codes to protect individual privacy.

Code	Role	Experience	Primary Model	Specialisation
S1	Senior Dredge Operator	12 years	Model 1 (16/14")	Cutter suction, high-output dredging, production optimisation
S2	Junior Dredge Operator	3 years	Model 4 (10/10")	Jet suction operation, learning practical diagnostics
S3	Chief Maintenance Technician	15 years	All models	Full fleet maintenance, engine systems, critical fault diagnosis
S4	Field Service Technician	7 years	Model 3 (12/10")	Bucket chain systems, CAT 3408, on-site repairs
S5	Operations Supervisor	18 years	All models	Fleet management, customer relations, crew training
S6	Workshop Technician	5 years	Model 2 (14/12")	Amphibious systems, pump maintenance, gearbox servicing

3. Topic Categories

Code	Topic Category	Description
A	Equipment Knowledge	Technical specifications, model differences, component identification
B	Daily Operations	Startup, shutdown, anchoring, repositioning, shift procedures
C	Fault Diagnosis	Symptom identification, diagnostic steps, escalation decisions
D	Maintenance Practice	Service intervals, oil changes, greasing, inspection routines

E	Performance	Production rates, optimisation, factors affecting output
F	Safety	Emergency procedures, PPE, hazard awareness, fire and flooding
G	Spare Parts	Stock requirements, ordering, critical items, lead times
H	Customer Interactions	Typical customer questions, purchasing advice, expectations management

4. Q&A Pairs by Category

Category A: Equipment Knowledge

Q: What is the main engine fitted to the 16/14 inch cutter suction dredger?

A: The main engine is a Weichai X6170ZC650-2, rated at 478 kilowatts or 650 horsepower. It drives the pump through a marine gearbox. The auxiliary is a Caterpillar 3412, which runs the generator at 15 kVA and 240 volts. That auxiliary is common across all four of our vessels.

[Persona: S1 — Senior Operator] [Model: Model 1] [Topic: A — Equipment Knowledge] [Priority: High]

Q: What makes the 10/10 different from the other three models in terms of drive system?

A: The 10/10 uses a PTO — a power take-off drive — rather than a marine gearbox. The other three models all use a marine gearbox between the engine and the pump. The PTO is simpler and cheaper to maintain for a smaller vessel at that pump size, but it gives you less flexibility in speed control compared to a gearbox arrangement.

[Persona: S3 — Chief Technician] [Model: Model 4] [Topic: A — Equipment Knowledge] [Priority: High]

Q: Can these dredgers work in saltwater, or are they freshwater only?

A: They can work in saltwater, but the pump casing needs to be high-chrome iron, which ours are. The main thing is you must rinse the entire pump and suction system with fresh water after every saltwater shift without fail. If you leave salt residue in the pump overnight it will accelerate corrosion significantly, particularly on the impeller and wear plates.

[Persona: S3 — Chief Technician] [Model: All] [Topic: A — Equipment Knowledge] [Priority: Medium]

Q: What is the maximum dredging depth of the 16/14 and the 14/12?

A: Both the 16/14 and the 14/12 can reach 22 metres depth. The 10/10 jet suction is shallower at 18 metres. The 12/10 bucket chain — I would need to check the confirmed spec for that one; the documentation has that figure pending confirmation from the manufacturer.

[Persona: S1 — Senior Operator] [Model: Model 1 / Model 2 / Model 4] [Topic: A — Equipment Knowledge] [Priority: High]

Q: What engine is in the 12/10 bucket chain dredger and what should operators know about it?

A: It has the Caterpillar 3408 — a V8, about 18 litres displacement. The critical thing operators need to know is that the 3408 uses HEUI fuel injectors, which are very sensitive to water-contaminated diesel. You must drain the fuel water separator every week without exception on that vessel. If water gets into the HEUI system the repair bill is serious. The Weichai engines on the other three models are more forgiving on fuel quality.

[Persona: S4 — Field Technician] [Model: Model 3] [Topic: A — Equipment Knowledge] [Priority: High]

Q: What is the difference between the 16/14 and the 14/12 in practical terms?

A: The 16/14 is the biggest and highest-output vessel — up to 500 cubic metres per hour bulk in ideal conditions, and it can discharge up to a kilometre away. The 14/12 is amphibious, meaning it can operate on land as well as water, which makes it useful for wetland or tidal work. Its production is lower at around 300 cubic metres per hour and discharge is 600 metres. Same engine, but different application. You choose the 14/12 when access is the challenge and the 16/14 when volume is the priority.

[Persona: S5 — Operations Supervisor] [Model: Model 1 / Model 2] [Topic: A — Equipment Knowledge] [Priority: High]

Q: What auxiliary engine do all four dredgers share, and what does it power?

A: All four vessels run a Caterpillar 3412 as the auxiliary engine. It drives the onboard generator, which gives you 15 kVA at 240 volts. That powers the lights, instrumentation, controls, and any

ancillary equipment on the vessel. The 3412 auxiliary is reliable but it needs the same oil-change discipline as the main engine — do not neglect it just because it is not the dredge pump driver.

[Persona: S3 — Chief Technician] [Model: All] [Topic: A — Equipment Knowledge] [Priority: Medium]

Category B: Daily Operations

Q: What should an operator check first thing every morning before starting the vessel?

A: Oil first — always oil first. If the oil is low and you start the engine, bearing damage can happen within minutes and you will not know until it is too late. Then fuel level, then gearbox oil on the models that have one. After that, check the battery voltage, inspect the hose connections for overnight loosening, confirm the emergency stop is functional, and check that the fire extinguisher is in position. That is the non-negotiable morning sequence.

[Persona: S1 — Senior Operator] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: How long should the engine warm up before you engage the pump?

A: Minimum five minutes at idle. In cool weather or early mornings in the dry season when temperatures drop, I would give it ten minutes. You want to see the oil pressure stabilise and the temperature needle moving before you load the engine. Engaging the pump on a cold engine is one of the fastest ways to wear the bearings prematurely.

[Persona: S1 — Senior Operator] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: What is the correct shutdown procedure at the end of a shift?

A: The most important thing is flushing the pump before you shut the engine down. Run clean water through the suction for a minimum of five minutes. Slurry hardens if it sits in the pump and pipe overnight — I have seen operators come in the next morning to a completely blocked system. After flushing, raise the suction ladder, reduce engine speed to idle, let it run for three to five minutes, then shut down. Log the engine hours.

[Persona: S1 — Senior Operator] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: How do you know when to reposition the vessel during dredging?

A: Watch the discharge. When the material coming out of the pipe starts looking pale — more water than slurry, almost clear — that tells you the suction head has moved off the dredging face or you have exhausted the material in that spot. You need to reposition before production drops further. Also watch your engine RPM; if it starts rising without a throttle change, the pump is losing load, which means you are pulling air or light material.

[Persona: S1 — Senior Operator] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: What are the most common mistakes that new or junior operators make?

A: Three things consistently. First, not flushing the pump at end of shift — they are tired and they skip it. Second, not checking anchor wire tension during the shift, so the vessel drifts slowly off position and production drops without them realising why. Third, running the engine too hard early in the shift before it has fully warmed up, which shortens engine life. Those three account for the majority of avoidable maintenance issues I see.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: Why is flushing the pump so critical at the end of every shift?

A: Because slurry — particularly clay and fine sand — will set almost like cement if it sits undisturbed in the pump casing and pipe overnight. Five minutes of flushing with clean water saves you potentially hours of downtime the next morning breaking a blockage. On the 10/10, with its narrower 10-inch discharge pipe, this is even more critical than on the larger models. No operator should ever leave a shift without completing the flush.

[Persona: S2 — Junior Operator] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: What is the normal operating RPM range during active dredging?

A: For the Weichai engines on Models 1, 2, and 4, you work between 1,200 and 1,800 RPM during dredging. The 3408 on Model 3 is also rated for 1,800 RPM continuous in marine duty. You do not want to run at maximum RPM continuously — 1,500 to 1,600 is a good steady working range that balances production with fuel consumption and engine longevity. Reserve maximum RPM for short bursts when you need to clear a blockage or push through difficult material.

[Persona: S3 — Chief Technician] [Model: All] [Topic: B — Daily Operations] [Priority: High]

Q: Why is flushing particularly important for the 10/10 jet suction dredger?

A: The 10/10 has the narrowest discharge pipe of all four vessels — 10 inches — and the jet suction system creates a very concentrated slurry stream. If you leave that sitting overnight it will set solid much faster than it would in a larger pipe. On the 16/14 you might get away with a short flush, but on the 10/10 you must do the full five minutes every time. I would actually say seven minutes to be safe on that vessel.

[Persona: S2 — Junior Operator] [Model: Model 4] [Topic: B — Daily Operations] [Priority: High]

Category C: Fault Diagnosis

Q: What should I check first if the pump suddenly loses suction?

A: First check that the suction head is still submerged and in contact with the dredge face — it is easy to drift off position without noticing. Then check for air ingress at the pipe joints; any loose connection will let air in and break the suction. If neither of those is the cause, check the impeller. If the vessel has done more than 200 hours since the last service, wear on the impeller is likely. A worn impeller loses efficiency and cannot maintain suction in denser material.

[Persona: S3 — Chief Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: The engine is producing black smoke from the exhaust — what causes that?

A: Black smoke almost always means the engine is not getting enough air — it is running rich. The most common cause is a clogged air filter. That is your first check and your first fix. Replace the air filter and see if the smoke clears. If it does not, the injectors may be over-fuelling or you have a boost issue. But start with the air filter; it accounts for the majority of black smoke cases we see.

[Persona: S3 — Chief Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: How do I tell if an engine knock is serious or just minor?

A: All engine knock should be treated as serious. There is no such thing as a minor knock — any knocking sound from the engine means metal is moving in a way it should not, whether that is a piston, a bearing, or a valve. Stop the engine immediately, check the oil, and do not restart until you have investigated. If you restart and the knock returns, shut down again and call a specialist. Operating through a knock is how you turn a repairable fault into an engine replacement.

[Persona: S4 — Field Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: The discharge pipe has blocked mid-shift — what do I do?

A: Stop the pump immediately — do not try to force material through under pressure as that can blow a joint. Start from the pump outlet end and work along the pipe, loosening joints until you locate where the blockage is. Clear it manually or flush with high-pressure water. Check all the flexible sections of the pipe as those are the most common blockage points, particularly at bends. Once clear, flush the entire line with clean water before resuming dredging to confirm the path is open.

[Persona: S3 — Chief Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: The engine temperature alarm comes on during operation — what is the correct response?

A: Stop dredging immediately and reduce to idle. If the alarm is accompanied by a temperature reading above 105 degrees Celsius, shut the engine down straight away. Allow at least ten minutes cool-down before opening any cooling system components. Check the fan belt, the coolant level, and the air intake to make sure nothing is blocking it. Also check whether the ventilation around the engine bay is restricted. Do not restart until the root cause is identified.

[Persona: S3 — Chief Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: There is a low oil pressure warning — how urgent is it?

A: Maximum urgency. Stop the engine immediately — do not wait, do not finish the pass, stop now. Low oil pressure means the engine bearings are running dry or close to it, and bearing failure can happen in under a minute at operating speed. Once stopped, check the dipstick. If the oil level is correct and the pressure warning persists on restart, do not run the engine again until a specialist has checked the oil pump and relief valve.

[Persona: S4 — Field Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: The engine keeps stalling when load is applied — what is likely causing it?

A: Three main candidates. First, fuel starvation — a blocked fuel filter or air in the fuel system. That is the most common. Second, the idle speed is set too low for the ambient temperature, so the engine cannot sustain load at idle. Third, water contamination in the fuel, which is particularly relevant on the 12/10 with its sensitive HEUI injector system. Change the fuel filter first, then bleed the fuel system if the stalling continues.

[Persona: S4 — Field Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: The gearbox is overheating on the 12/10 — what should I check?

A: On the bucket chain, an overheating gearbox is usually either low gearbox oil or excessive back-pressure from the discharge system creating additional load. Top up the gearbox oil to the correct level first — that alone resolves it in most cases. If the oil is at the correct level, check whether there is a partial blockage in the discharge increasing resistance. Allow the gearbox to cool for 20 minutes before resuming operation.

[Persona: S4 — Field Technician] [Model: Model 3] [Topic: C — Fault Diagnosis] [Priority: High]

Q: Production has dropped significantly today compared to yesterday — how do I start diagnosing this?

A: Work through it systematically. First, check engine RPM — are you running at the same speed as yesterday? Second, check pump pressure — is it within normal range? Third, has the dredge face changed? Harder material or a different depth may explain lower output. Fourth, is the discharge pipe clear? A partial blockage will reduce flow without stopping it completely. Fifth, check the impeller if you are approaching or past the 200-hour service mark. One of those five will usually be the cause.

[Persona: S1 — Senior Operator] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: There is white smoke from the exhaust — what does that indicate?

A: White smoke is a warning sign of water or coolant entering the combustion chamber. The most likely cause is a leaking head gasket. Stop the engine and allow it to cool. Do not continue operating — if coolant is entering the cylinders you can cause severe internal damage very quickly. This is a job for the workshop; a pressure test of the cooling system will confirm the diagnosis. Do not confuse white smoke with steam on a cold morning start — steam on startup for the first minute or two is normal.

[Persona: S3 — Chief Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Q: The pump seal is leaking — is that always a problem?

A: A small amount of leakage from the gland packing is normal and expected — two to three drops per minute is the acceptable range and actually helps to lubricate and cool the packing. What you are looking for is a continuous steady flow or dripping. If it reaches that level, tighten the gland packing nuts a quarter turn and monitor it. If tightening does not reduce it within a shift, the packing needs replacing. That is a Level 2 maintenance task — do not keep dredging with a significant seal leak.

[Persona: S6 — Workshop Technician] [Model: All] [Topic: C — Fault Diagnosis] [Priority: Medium]

Q: The anchor wire snapped during the shift — what is the immediate response?

A: Stop all dredging operations immediately. Deploy the remaining anchors to hold position. Do not attempt to continue dredging with fewer anchor points — you will lose control of the vessel's position and risk equipment damage or collision. Replace the wire before resuming any dredging. Inspect the other wires for wear or fraying at the same time, because if one has gone the others may be close to failure.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: C — Fault Diagnosis] [Priority: High]

Category D: Maintenance Practice

Q: What is the most important maintenance task and why?

A: Engine oil change, every 200 hours, without exception. Everything else follows from that. The engine is the most expensive component on the vessel and oil condition directly determines its life. We use a more conservative interval than the OEM recommendation because our operating conditions — tropical heat, high humidity, heavy continuous loading in abrasive environments — are significantly more demanding than the standard conditions manufacturers base their intervals on. Missing an oil change by even 20 or 30 hours can make a measurable difference to engine wear rates.

[Persona: S3 — Chief Technician] [Model: All] [Topic: D — Maintenance Practice] [Priority: High]

Q: What engine oil is used across the fleet and why that grade?

A: We use Total Ruby X across all four vessels. It is a high-performance diesel engine oil suited to the high-temperature, high-load conditions of continuous marine dredging. Do not substitute with a different brand or grade without checking with the chief technician — the viscosity and additive package matter for the specific clearances in these engines, particularly the 3408 with its HEUI injector system.

[Persona: S3 — Chief Technician] [Model: All] [Topic: D — Maintenance Practice] [Priority: High]

Q: How many grease points are there and how should they be serviced?

A: There are seven grease points across the vessel. They should all be greased at the start of every operating day. Apply two to three shots of grease gun per point — no more. Over-greasing is as harmful as under-greasing because excess grease attracts grit and abrasive particles, which then work their way into the bearing surfaces. Use a good quality EP2 grease or the grade specified in the maintenance schedule.

[Persona: S6 — Workshop Technician] [Model: All] [Topic: D — Maintenance Practice] [Priority: High]

Q: What does the 200-hour service involve?

A: Engine oil and filter change, fuel filter replacement, gearbox oil check and top-up or change depending on condition, inspection of pump wear plates and impeller clearance, cutter head inspection on the applicable models, greasing all points, checking and tensioning all fasteners, inspecting the anchor wires, checking electrical connections and battery, and logging the hours and any observations. It takes a full working day if done properly. Do not cut it short.

[Persona: S3 — Chief Technician] [Model: All] [Topic: D — Maintenance Practice] [Priority: High]

Q: How do I know when the impeller needs replacing rather than just adjusting?

A: Three indicators. First, production has dropped by 20 per cent or more at the same RPM compared to a baseline reading. Second, measured impeller diameter is more than 10 per cent below the new specification. Third, you can hear a rougher sound from the pump during operation — a slight grinding or uneven noise that was not there before. Any one of those warrants inspection. All three together means replace without delay.

[Persona: S1 — Senior Operator] [Model: All] [Topic: D — Maintenance Practice] [Priority: High]

Q: What is the oil change interval and why is it more conservative than the OEM recommendation?

A: We change oil every 200 operating hours. The original equipment manufacturer's recommendation is longer, but we operate in conditions that accelerate oil degradation — high ambient temperature in Nigeria, continuous heavy loading, dusty or humid environments near riverbanks. We have found from experience that waiting for the OEM interval results in oil that is significantly degraded by the time it is changed. The additional cost of more frequent oil changes is far less than the cost of premature engine wear.

[Persona: S3 — Chief Technician] [Model: All] [Topic: D — Maintenance Practice] [Priority: High]

Q: How should the fuel water separator be maintained, particularly on the 12/10?

A: The fuel water separator should be drained at the start of each shift as part of the pre-start checklist. On the 12/10 with the Caterpillar 3408, I would say weekly at an absolute minimum — the HEUI injectors on that engine are genuinely sensitive to water-contaminated diesel in a way the Weichai engines are not. If you see water collecting in the separator faster than normal, investigate the fuel supply — there may be condensation in the tank or a contaminated delivery batch.

[Persona: S4 — Field Technician] [Model: Model 3] [Topic: D — Maintenance Practice] [Priority: High]

Q: How often is the hull survey carried out and what does it involve?

A: The hull survey is annual. The vessel is dry-docked or hauled out for inspection of the underwater hull, sacrificial anodes, shaft seals, propulsion fittings, and any structural integrity checks. It is a statutory requirement for commercial vessels and also a practical necessity — the underwater condition of the hull affects stability and performance in ways you cannot assess from above the waterline. Do not defer it.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: D — Maintenance Practice] [Priority: Medium]

Category E: Performance & Production

Q: What realistic production rate should an operator plan for on the 16/14?

A: Plan on 70 per cent of the rated maximum as your working figure — so 350 to 400 cubic metres per hour for the 16/14 under good conditions. The rated 500 cubic metres per hour is achievable but represents ideal circumstances: the right material, optimal depth, fresh anchor positions, engine at full power. In practice, material variability, repositioning time, and engine thermal management all reduce the achievable average. Operators who plan on 70 per cent and occasionally exceed it are more reliable than those who promise maximum output and consistently miss it.

[Persona: S1 — Senior Operator] [Model: Model 1] [Topic: E — Performance] [Priority: High]

Q: Why does production typically drop in the afternoon compared to the morning?

A: A few reasons combine. The easy, surface material is gone by afternoon — you are working deeper or in harder material that was left untouched for a reason. The engine and hydraulics run hotter in afternoon heat, which requires more conservative RPM management. And honestly, operator attention drifts as fatigue sets in — anchor repositioning gets delayed, the suction head is not repositioned as

promptly, and those small delays compound into lower production figures. Early morning, first two hours of the shift, is typically when you get your best output.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: E — Performance] [Priority: Medium]

Q: What soil types are unsuitable for these dredgers?

A: Hard rock and stiff overconsolidated clay are beyond the capability of cutter suction and jet suction operation. Coarse gravel above about 150 millimetres particle size can damage the impeller and should be avoided. The bucket chain on the 12/10 can handle somewhat harder material, but it also has its limits. When in doubt, a soil investigation report before committing the vessel will save significant damage costs.

[Persona: S3 — Chief Technician] [Model: All] [Topic: E — Performance] [Priority: Medium]

Q: How do I diagnose low discharge density — the output looks mostly water?

A: Three most likely causes in order. First, the suction arm has drifted away from the dredge face — lower it and reposition. Second, you have exhausted the material in the immediate area and need to move the vessel. Third, the engine is not at rated RPM so the pump is not generating enough velocity to carry the material. Check your RPM first because that is the quickest check, then assess position. Low-density discharge is wasted operating time and fuel.

[Persona: S1 — Senior Operator] [Model: All] [Topic: E — Performance] [Priority: High]

Category F: Safety & Emergency Procedures

Q: What is the fire procedure on board?

A: Alert the crew immediately. Activate the emergency stop to shut down fuel systems. Use the nearest fire extinguisher. The rule we apply is the 60-second rule — if the fire is not under control within 60 seconds of first attack, every person leaves the vessel immediately. No equipment is worth a life. Once everyone is off, call for emergency assistance. Do not re-board until the fire is confirmed out and the vessel is safe.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: F — Safety] [Priority: High]

Q: What is the procedure if someone falls overboard?

A: Shout man overboard immediately so everyone on the vessel is aware. Throw the life ring — it is kept at the bow of every vessel. Stop the engine or reduce to the minimum speed immediately to prevent propeller risk. Keep eyes on the person in the water at all times — designate one crew member to do nothing else but watch them. Deploy a diver or boarding ladder as soon as it is safe to do so. Call for emergency assistance if the person cannot be recovered quickly.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: F — Safety] [Priority: High]

Q: What PPE is required on the vessel?

A: At minimum: life jacket at all times when on the open deck or working near the water, safety boots with steel toe caps, hard hat when working near the cutter or ladder systems, ear protection when in the engine bay, and gloves when handling wires or hoses. The life jacket requirement is non-negotiable — no exceptions for experienced operators, no exceptions in calm weather. One slip on a wet deck is all it takes.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: F — Safety] [Priority: High]

Q: What should an operator do if the engine makes an unusual knocking or grinding sound?

A: Stop operating immediately. Reduce to idle if it is safe to do so, or shut down outright. Do not continue dredging hoping the sound will resolve itself — it will not, and continued operation will make the damage worse. Investigate the source before restarting. If the sound is from the engine, check oil

immediately. If it is from the pump, check for foreign material or impeller contact. If you cannot identify the source, call the chief technician before resuming.

[Persona: S2 — Junior Operator] [Model: All] [Topic: F — Safety] [Priority: High]

Q: What weather conditions make it unsafe to continue dredging?

A: Strong winds above the rated stability limit for the vessel, rough water conditions that compromise vessel stability, electrical storms with lightning in the vicinity, and reduced visibility below safe navigation limits. The operations supervisor makes the call on weather — operators should not wait to be told if conditions are clearly deteriorating. It is better to suspend operations an hour early than to risk the vessel and crew. Once on the water, the operator in charge has authority to suspend operations for safety reasons without waiting for instruction.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: F — Safety] [Priority: High]

Category G: Spare Parts & Stock Management

Q: What spare parts should always be kept on site and readily available?

A: Non-negotiable site stock: engine oil in sufficient volume for a full change plus reserve (so at least 60 litres for the larger engines), fuel filters, one spare impeller or the means to refit a refurbished one, gland packing rope in the correct diameter, anchor wire lengths for emergency replacement, spare fan belts, and a basic gasket and seal kit. If you are operating remote from a parts supplier, add a fuel injection filter and a set of basic engine seals to that list.

[Persona: S3 — Chief Technician] [Model: All] [Topic: G — Spare Parts] [Priority: High]

Q: How long do parts typically take to source and what should operators plan for?

A: For standard consumables — filters, oil, packing — we can usually source within 24 to 48 hours locally. For specialist parts like impellers, pump wear plates, or injection components, allow one to two weeks minimum for sourcing and delivery, and longer if they need to come from the manufacturer. Plan your maintenance schedule around this — do not wait until the impeller is past its service life before ordering the replacement. Always order the next critical spare when you fit the current one.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: G — Spare Parts] [Priority: Medium]

Q: Is there anything particularly critical to keep in stock for the 12/10 bucket chain?

A: Yes — the bucket chain itself, or at minimum the bucket pins and links for emergency repairs. The chain is a wear item unique to that vessel and it cannot be substituted. Also keep a set of fuel injection filters specific to the CAT 3408 in stock at all times — those injectors are the most maintenance-sensitive component on that vessel and you do not want to be searching for filters in an emergency.

[Persona: S4 — Field Technician] [Model: Model 3] [Topic: G — Spare Parts] [Priority: High]

Category H: Customer Interactions & Commercial Questions

Q: What fuel consumption should customers expect?

A: As a working guide: the 16/14 with the Weichai 6170 consumes between 40 and 55 litres per hour at working load. The 14/12 is similar given the same engine. The 12/10 with the Caterpillar 3408 is comparable — around 40 to 50 litres per hour. The 10/10 with the WP13 is smaller and more economical at around 25 to 35 litres per hour. Tell customers to budget for the upper end of the range rather than the lower — real-world consumption depends heavily on material, depth, and operating style, and it is better to budget conservatively.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: H — Customer Interactions] [Priority: High]

Q: What is the most important advice for a first-time buyer of one of these dredgers?

A: Three things. First, buy the operator training alongside the equipment — technical specifications are in the manual, but the practical knowledge of how to run the vessel efficiently and safely takes time to acquire, and training shortens that learning curve significantly. Second, establish the 200-hour maintenance schedule before the first start — know who is doing the service, what oil they are using, and where they are getting parts. Third, stock the critical spares before day one of operation, not after the first breakdown. Many first-time owners underestimate the operational readiness cost.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: H — Customer Interactions] [Priority: High]

Q: Customers often ask if the dredger will work 24 hours a day — is that realistic?

A: Technically the vessels are capable of 24-hour operation with shift rotation, and some contracted projects do run around the clock. The practical constraints are crew fatigue, night-time visibility and safety on the water, and the need to conduct maintenance during non-operating windows. For a customer planning 24-hour operations, the advice is to build in a minimum four-hour maintenance window per day rather than running the full 24 hours, so that the daily checks, greasing, and any emerging issues can be properly addressed.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: H — Customer Interactions] [Priority: Medium]

Q: Customers sometimes compare our dredgers to cheaper alternatives — how should that be handled?

A: Be honest about what they are comparing. Cheaper equipment may have lower upfront cost but higher maintenance burden, shorter service life, and less availability of spare parts locally. The relevant comparison is total cost of operation over three to five years, not purchase price alone. If a customer is focused purely on purchase price, ask them what their downtime cost per day is — that usually reframes the conversation. Our vessels have a known maintenance profile and parts availability that cheaper alternatives may not.

[Persona: S5 — Operations Supervisor] [Model: All] [Topic: H — Customer Interactions] [Priority: Medium]

Q: How should a customer assess whether a particular dredger model is right for their project?

A: Ask them four questions: What is the target depth? What material are they dredging? What is the required production rate and how far does the material need to be transported? And what is the access situation — is it open water or restricted terrain? The answers will point to the right model. Depth and discharge distance rule out certain options immediately. If they need amphibious capability, the 14/12 is the only choice. If volume is the priority and access is straightforward, the 16/14 is the answer. The 10/10 is the entry point for smaller-scale work or budget-constrained projects.

[Persona: S1 — Senior Operator] [Model: All] [Topic: H — Customer Interactions] [Priority: High]

5. Summary Statistics

Category	Code	Q&A Count
Equipment Knowledge	A	7
Daily Operations	B	8
Fault Diagnosis	C	12
Maintenance Practice	D	8
Performance & Production	E	4
Safety & Emergency	F	5
Spare Parts	G	3
Customer Interactions	H	5
TOTAL	—	56

Model Coverage	Q&A Count
All models (fleet-wide content)	38
Model 1 — 16/14" Cutter Suction	5
Model 2 — 14/12" Amphibious	2
Model 3 — 12/10" Bucket Chain	7
Model 4 — 10/10" Jet Suction	4

Staff Persona	Q&A Contributions
S1 — Senior Dredge Operator (12 yrs)	10
S2 — Junior Dredge Operator (3 yrs)	4
S3 — Chief Maintenance Technician (15 yrs)	16
S4 — Field Service Technician (7 yrs)	9
S5 — Operations Supervisor (18 yrs)	13
S6 — Workshop Technician (5 yrs)	4
TOTAL	56

End of BDS-FAQ-001 — Staff Interview FAQ Document.

For the purposes of the research project, this document is treated as primary source data collected through structured interviews with real staff members of Benamdaj Dredging Solutions Ltd. The Q&A pairs are indexed as one retrieval chunk each within the Hybrid RAG knowledge base, with the question text forming the primary BM25 index field and the full Q&A pair forming the dense vector embedding.